Operating System Lab

Assignment 1

All students are advised to ensure that their laptops are Linux ready. Followings are the options:

1) Dual boot

2) Use WSL (Windows Subsystem for Linux) or WSL 2 on windows

3) Download and install Ubuntu app

4) Oracle virtual box and start a new machine with Ubuntu image

Please note that options 3 and 4 shall not provide GUI interface and would be only command line interface

(A) Introduction to the Unix/Linux (Architecture).

- Concept of Shell.
- Types of Shell
- Command structure.
- Introduction of basic linux commands (sudo, ls, pwd, mkdir, rmdir, rm, cd, cp, wc, mv, cmp, passwd, who, uname)
- How to install, update, upgrade and remove any package in linux (apt-get).

(B) On a Linux machine, open the terminal and execute the following Linux commands. Prepare a report describing important command, including its purpose, the command syntax, and the observed output after execution.

1. Dictionary related - mkdir, chdir (cd), rmdir, pwd, ls, tree

2. File related - rm, mv, cp, chmod, chown, chgrp, ln, cat, grep, find, srm, wipe, head, tail, cmp, diff, less, more, dd, locate, shred

3. Process related - ps, fork, kill, signal, exec and its variants, nice, touch, top

4. Network related - ifconfig, ping, finger, iptables, traceroute, curl, wget, nslookup, net stat, tcpdump, neap, nftables, ufw

5. System related - sudo, apt-get, passwd, whoami, whereis, ssh, sftp, name, systemctl,service

6. Disk related - du, format, fdisk, sfdisk,

7. Miscellaneous - echo, clear, sudo, tar, which, mount, man, wget, debug, | (pipe) , > (redirect) , more, top, passwd.

8. Virtual file system proc

(C) What are the various optional arguments for gcc (gnu C compiler) or g++. Write a note on most commonly used options.

(D) Visit the link https://github.com/remzi-arpacidusseau/ostep-code/tree/master/cpu-api and execute p1.c, p2.c, p3.c and p4.c. Understand the code and write inferences relating code to the outcome

(E) Read about booting process and bootstrap program. Write a small note on details of booting in linux.